Sets, relations, functions, and proofs

Michael Franke

Basic notions of (naïve) set theory; sets, elements, relations between and operations on sets; relations and their properties; functions and their properties. Examples of informal proofs: direct, indirect and counterexamples.

1 Naïve set theory

1.1 Sets, elements, universe

A *set* is a collection of entities. We use notation with curly braces "{. . .}" to represent such a collection. If we have entities a and b, examples of sets are:

$$X = \{a\} \qquad\qquad Y = \{a, b\}$$

The entity a is an *element* of X and Y. We write this as $a \in X$ and $a \in Y$. The entity b is not an element of X. We write this as $b \notin X$.

A set is individuated by the elements it contains. This means that the order of representation of elements is irrelevant. For example, $\{a, b\} = \{b, a\}$. This also means that whenever any two sets (however obtained) contain the same elements, they are identical. In other words, for any two sets X and Y to be different, there has to be at least one element $x \in X$ such that $x \notin Y$ or some $y \in Y$ such that $y \notin X$.

It is possible for a set to have no element at all. This set is called the *empty set* and we refer to it with the symbol $\emptyset$.

Occasionally we might wish to specify the *universe* U of all entities which are under consideration.[1] Any specification of a set is then implicitly restricted to entities in U.

§ 1

The Conception of Power or Cardinal Number

By an "aggregate" (*Menge*) we are to understand any collection into a whole (*Zusammenfassung zu einem Ganzen*) M of definite and separate objects m of our intuition or our thought. These objects are called the "elements" of M.

Figure 1: Passage first introducing the intuitive notion of a set from (the English translation of) Georg Cantor's *Beiträge zur Begründung der transfiniten Mengenlehre* from 1915.

[1] U need not be a set itself; it can be something bigger. But that is best left aside here. The whole concept of a universe might seem confusing at first sight. It is possible not to deeply understand what it is good for in the greater scheme of things, and still understand everything of current relevance about naïve set theory.

1.2 Ways of describing or defining sets

Three main methods for describing or defining sets exist:

1. by listing elements
2. by characteristic property
3. by recursive definition

The text above already gave examples for describing sets by *listing elements*. Sometimes we use notation "…" to indicate a range of elements when there is a clear intuitive ordering relation among them. Or we use "…" to abbreviate the obvious other members, even if there is no natural ordering. For example:

$X = \{2, 4, 6, 8, \dots\}$ [set of even integers bigger than 0]

$Y = \{2, 4, 6, 8, \dots, 20\}$ [set of even integers no bigger than 20]

$Z = \{\text{Russell}, \text{Wittgenstein}, \text{Frege}, \dots\}$[2] [set of authors to read]

[2] This would only be a good definition if the way to fill the "…" was absolutely clear.

To describe sets by a *characteristic property* of its elements, we might write:

$X = \{x \mid x \text{ is an even integer}\}$

$Y = \{x \mid x \text{ is an even integer no bigger than 20}\}$

$Z = \{x \mid x \text{ is a famous logician}\}$

To narrow down a reference set explicitly, we would write:[3]

$X = \{x \in \{1, 2, 3, \dots\} \mid x \text{ is even}\}$

$Y = \{x \in \{1, 2, 3, \dots, 20\} \mid x \text{ is even}\}$

$Y = \{x \in \{y \mid y \text{ is a logician}\} \mid x \text{ is famous}\}$

[3] With a universe U in place, we should read a description like

$$X = \{x \mid \text{property of } x\}\,, \quad \text{as}$$
$$X = \{x \in U \mid \text{property of } x\}\,.$$

To describe sets by *recursive definition*, we must:[4]

(i) anchor the recursion

(ii) specify a recursion step

(iii) exclude elements untouched by anchor or recursive steps

[4] Recursive definitions are useful because they allow for easier proofs and easier further definitions. This will become clear when we look at a recursive definition of the formulas of a logical language, to which we will then assign a meaning by exploiting the original recursive definition (keyword: "Tarski truth conditions").

An example is the following definition of natural numbers:

(i) 0 is a natural number

(ii) if n is a natural number, then so is $n + 1$

(iii) nothing else is a natural number

Another example is the definition of a simple formal language $\mathfrak{L}$. Unlike the previous example, we here define a set of symbols:[5]

1. words for all natural numbers are elements of $\mathfrak{L}$ (e.g., "one", "two", …)
2. if $x, y \in \mathfrak{L}$, then so are the strings:

   "x plus y" "x minus y"

3. no string which is not constructible by this procedure is in $\mathfrak{L}$

[5] Examples of elements of $\mathfrak{L}$ by this definition are: "three", "twenty minus three", "twenty minus three plus four". Not an element of $\mathfrak{L}$ are "minus three", "plus one two", "one plus minus two".

1.3 Important numerical sets to be familiar with

Important sets to be familiar with are:[6]

$\mathbb{N} = \{0, 1, 2, 3, \dots\}$ [set of natural numbers]

$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ [set of integers]

$\mathbb{Q} = \{p/q \mid p \in \mathbb{Z}, q \in \mathbb{N} \setminus \{0\}\}$ [set of rational numbers]

$\mathbb{R} = \{\pi, 0, e, \dots\}$ [set of real numbers]

[6] Notice the commonly used "double-stroke notation" for the capital letters used to refer to these special sets.

1.4 Cardinality

The number of elements in a set is called its *cardinality*. We write $|X|$ for the cardinality of X. The cardinality of X can be infinite. We then write $|X| = \infty$ and say that X is an infinite set.[7] If X is not an infinite set, it is called a *finite set*. Examples:

$$|\{a\}| = 1 \qquad |\{a, b\}| = 2$$
$$|\emptyset| = 0 \qquad |\{2, 4, 6, 8 \dots\}| = \infty$$

[7] Actually, infinite sets can have different cardinalities, so that writing $|X| = \infty$ could be misleading. For example, $|\mathbb{N}| = |\mathbb{Q}| < |\mathbb{R}|$. But this is not important for us at the moment.

1.5 Relations between sets

A set Y can contain another set X. Inversely, a set X can be an element of another set Y. We then write $X \in Y$. For example:

$X = \{a, b\}$

$Y = \{c, d, X\} = \{c, d, \{a, b\}\}$

It is important to note that $\{a, b\} \in Y$ but $a \notin Y$.

If all of the elements of X are also in Y, we say that X is a *subset* of Y, or that Y is a *superset* of X, and we write $X \subseteq Y$. If $X \subseteq Y$ and there is at least one element in Y which is not in X, we say that X is a *proper subset* of Y, or that Y is a *proper superset* of X, and we write $X \subset Y$. If X is not a (proper) subset of Y, we write $X \not\subseteq Y$ ($X \not\subset Y$). Some examples:

$\{a, b\} \subseteq \{a, b, c\}$ $\qquad$ $\{a, b\} \subset \{a, b, c\}$

$\{a, b\} \not\subseteq \{a, c\}$ $\qquad$ $\{a, b\} \not\subset \{a, b\}$

1.6 Operations on sets

Operations on sets take one or several sets as input and return another set. We consider here the power set operation and different kinds of logical operations.

The *power set* $\mathcal{P}(X)$ of X is the set of all subsets of X:

$$\mathcal{P}(X) = \{Y \mid Y \subseteq X\}$$

If X is finite, the cardinality of $\mathcal{P}(X)$ is $2^{|X|}$.[8] For example:

[8] This is because we decide $|X|$ times whether to include an element or not; so we collect all outcomes of $|X|$ binary decisions.

$X = \{a, b\}$ $\quad |X| = 2$

$\mathcal{P}(X) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$ $\quad |\mathcal{P}(X)| = 2^2 = 4$

The following operations on sets correspond to logical operators (*and*, *or*, *not*). For any sets X and Y:

$X \cap Y = \{z \mid z \in X \text{ and } z \in Y\}$ [intersection]

$X \cup Y = \{z \mid z \in X \text{ or } z \in Y\}$ [union]

$X \setminus Y = \{z \mid z \in X \text{ and } z \notin Y\}$ [difference]

$\overline{X} = \{z \in U \mid z \notin X\}$ [complement[9]]

Here are some examples:

$X = \{a, b, c\}$ $\quad X \cup Y = \{a, b, c, d\}$

$Y = \{b, c, d\}$ $\quad X \setminus Y = \{a\}$

$X \cap Y = \{b, c\}$ $\quad Y \setminus X = \{d\}$

A number of facts follows from the definitions so far. Some are shown in Figure 3. To conclusively show that something follows from a definition, we need the concept of a *proof*, the topic of the next section.

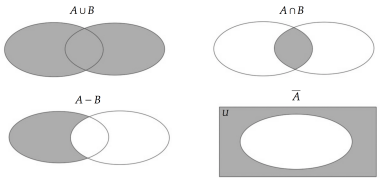


Figure 2: Venn diagrams of set operations.

[9] We need an explicit universe to interpret the complement operation.

1. Commutative Laws
   $X \cup Y = Y \cup X$ $\quad X \cap Y = Y \cap X$
2. Associative Laws
   $(X \cup Y) \cup Z = X \cup (Y \cup Z)$ $\quad (X \cap Y) \cap Z = X \cap (Y \cap Z)$
3. Distributive Laws
   $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$
   $X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$
4. Identity Laws
   $X \cup \emptyset = X$ $\quad X \cup U = U$ $\quad X \cap \emptyset = \emptyset$ $\quad X \cap U = X$
5. Complement Laws
   $X \cup X' = U$ $\quad (X')' = X$ $\quad X \cap X' = \emptyset$ $\quad X - Y = X \cap Y'$
6. DeMorgan's Laws
   $(X \cup Y)' = X' \cap Y'$ $\quad (X \cap Y)' = X' \cup Y'$

Figure 3: Some facts. NB: The complement of set X is written as X' in this list.

Exercise 1. Provide a natural language paraphrase for each of the following sets:

a. $A = \{5, 7, 9, 11, 13, \dots\}$

b. $B = \mathbb{N} \setminus A$

c. $C = \{c \in \mathbb{Z} \mid -2 \leq c \leq 2\}$

d. $D = \mathbb{N} \cap \overline{C}$

Exercise 2. Let's assume the following definitions (assuming that the universe is $U = X \cup Y \cup Z$):

$$X = \{a, b, c, d\}$$
$$Y = \{y \mid y \text{ is a vowel}\} = \{a, e, i, o, u\}$$
$$Z = \{z \mid z \text{ is an even natural number smaller than } 5\}$$

Write out the results of the following set operations:

a. $X \cap Y$

b. $X \cup Y$

c. $X \setminus Y$

d. $Y \setminus Z$

e. $Y \setminus Y$

f. $X \cap \overline{X}$

Exercise 3. For each of the following statements, determine whether it is true or false, using the sets X, Y and Z as defined above:

a. $X \subset Y$

b. $Y \in X$

c. $X \cap Y \subseteq X$

d. $|X| = |Y|$

e. $X \cup Z \subseteq X$

f. $X \cap Y \not\subset X$